



JACOB FOSTER

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Experience

Advanced Micro Devices (AMD)

May 2025 - August 2026

Graphics Software Engineer Intern

Markham, ON

- Engineered software for **compressing and decoding** MJPEG video streams via **DCT-based decoding**, enabling hardware acceleration for next-generation Windows GPU, APU, and console drivers across D3D11/12 decode paths.
- Built a bitstream parser that walks JPEG marker structure, detects chroma subsampling, and reformats frames into DXVA-compliant picture parameters and Huffman/quantization tables; enabling the team to validate hardware decoding for product bring-up.
- Designed and built a Python web app that automates the end-to-end firmware submission pipeline, unifying six previously manual stages and reducing submission time by **2.3x**. The tool orchestrates remote **Linux** test machines over **SSH**, drives **Perforce** and **Git** from a single interface, auto-generates and cross-links pull requests across three downstream repositories, and integrates an **AI coding agent** to resolve merge conflicts. Deployed to a team of **30+ developers**.
- Resolved a scheduler defect where all video decode jobs were incorrectly dispatched to the encoding engine; implemented **codec-aware engine routing** to direct workloads to the dedicated decode engine, with AV1 exempt and single-engine APU configurations handled separately, leveraging **GPU decode affinity** to defer engine assignment until hardware capabilities were known at runtime.
- Diagnosed and resolved defects in Windows GPU/APU driver software and implemented features across the full driver stack, **from kernel-level to user-level abstractions**.

L3Harris Technologies - Wescam Division

May 2024 - August 2024

Manufacturing Engineering Intern

Waterdown, ON

- Developed a software/firmware comparison web tool to capture current and previous build components of a WESCAM imaging device and automatically determine build status for repairs, **decreasing effective labour time from 20 mins to 1 min**. Built using **C#, .NET Framework, HTML, CSS**.

McMaster Solar Car Project

October 2024 - October 2025

Race Strategies Algorithm Developer

Hamilton, ON

- Developed responsive algorithms using **Python and scikit-learn** to compute optimal travel speed and conserve energy based on solar irradiance, geological, and social factors.

Education

McMaster University, Hamilton, ON

Sep. 2022 - May 2027

Bachelor of Mechatronics Engineering - Co-op

cGPA - 3.86/4.00

- Relevant Coursework:** Data Structures and Algorithms, Embedded System Design, Analog and Digital Circuits, Software Development and Design, Operating Systems, FPGA Design.
- Accolades: Faculty of Engineering Award of Excellence Scholarship 2022, Dean's Honours List 2022-2026.

Projects

AllergyPal | *Python, Tensorflow, HTML, CSS, Javascript, Flask*

September 2024

- Designed a machine-learning driven website that allows users to input a list of dietary restrictions to **detect over 100 different food-based hazards**.
- Trained an image recognition model** to identify meals and ingredients, which was then cross referenced with a user's dietary restrictions.
- Created a **REST API using Flask and Python** to call upon the image recognition model in real time upon a user's request.

Spinnick SumoBot | *Arduino, C++, Autodesk Inventor, Embedded Systems, Soldering*

January 2024

- Developed C++ control algorithms for attack and evasion for an automated battle robot capable of perceiving environment features.
- Assembled the bot's control mechanism using an **Arduino micro-controller, ultrasonic and colour sensors, and DC motors**.

Technical Skills

Computing: Python, C, C++, C#, Verilog, ffmpeg, Microsoft D3D, Make, HTML, CSS, Javascript, React.js

Developer Tools: Git, GitHub, Perforce, Jira, Linux, WinDBG, .NET, Cursor, Claude CLI, VS Code, Visual Studio

Technologies: SOLIDWORKS, Autodesk Inventor, Adobe Photoshop, Microsoft Office

Manufacturing: Soldering, 3D printing, Drill press, Hand and power tools